SPECIMEN · VOLUME ONE



# Print*Craft* Showcase

## A tour of the page, glyph by glyph.

*Typography, world scripts, vector graphics, mathematics, forms and annotations — gathered in one document to put a PDF engine through its paces.*

**CLEONPDF**
Open-source PDF toolkit in Rust

Generated by `cargo xtask demo-pdf`
US Letter · 8.5 × 11 in

INSIDE THIS SPECIMEN

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### COLOPHON

Body text is set in EB Garamond with Playfair Display for display, Inter for labels, and JetBrains Mono for code. Every font is openly licensed (OFL) and listed in ATTRIBUTION.md. Layout is authored in HTML and CSS, printed by headless Chrome, then post-processed with `lopdf`.

### WHAT TO LOOK FOR

Open the bookmarks, layers, attachments and comments panels. Page labels run Cover, i–ii, then 1–10. One layer adds a draft watermark; another holds print-only notes and starts hidden.

FOREWORD

# Why a *specimen?*

A PDF reader is judged in the first second. The page appears, and either the letters sit on it the way the author intended or they do not. Every kerning pair, every ligature and every soft gradient is a small promise the file makes and the renderer must keep.

This document collects those promises in one place. It was written for the people who build CleonPDF, as a single file to open after every change: if all of it still looks right, the change is probably safe. It is also, we hope, pleasant to read on its own.

Nothing here is exotic. Each feature is something ordinary documents use every day — an invoice with a form field, a thesis full of equations, a contract with comments in the margin, a brochure with a translucent photograph laid over a tinted panel. What is unusual is only that they are all together, and that each page says plainly what it is testing.

The layout is authored in HTML and printed by a browser; the interactive parts are then written into the file by hand, object by object. That split mirrors the real world, where most PDFs are produced by one program and amended by several others.

*The CleonPDF contributors*

### TRACKING, FROM TIGHT TO AIRY

| | |
|---|---|
| MEASURE TWICE, SET ONCE | -40 |
| MEASURE TWICE, SET ONCE | 0 |
| MEASURE TWICE, SET ONCE | +120 |
| MEASURE TWICE | +300 |

### THE FAMILIES IN THIS SPECIMEN

| Ag | Ag | Ag | Ag | Ag | ∑x |
|---|---|---|---|---|---|
| **Playfair Display** | **EB Garamond** | **Inter** | **JetBrains Mono** | **Great Vibes** | **STIX Two Math** |
| Display | Body text | Labels | Code | Script | Mathematics |

CHAPTER ONE

# Setting *Text*

*Two justified columns with automatic hyphenation, old-style figures, a drop cap and a pull quote — the everyday machinery of a well-made page.*

Long before there were screens, a page of type was a negotiation between the width of a column and the length of the words that had to fill it. The compositor adjusted spaces, divided a word at a sensible syllable, and occasionally rewrote a line so the paragraph would lie flat. Justified text asks for exactly that care, and a renderer must honour the spacing it was given.[1]

A PDF does not reflow. Every glyph has already been placed by the program that wrote the file, and the reader's job is to reproduce those positions faithfully at any zoom. That sounds simple until one remembers that a single line may mix three fonts, a ligature standing in for two letters, and a hyphen that exists only because the line happened to break there in 1994 or 2026.

Old-style figures, like the ones in this paragraph's 1,280 characters and 36 lines, rise and fall with the lowercase so that numbers do not shout. Ligatures such as *fi*, *fl* and *ffi* in words like *officially*, *affluent* and *afflicted* join letters that would otherwise collide.[2]

> *"Good typography is noticed only when it is missing."*
>
> A WORKING MAXIM OF THE TRADE

Hyphenation is where browsers and word processors differ most. Chrome consults the system's hyphenation dictionaries when a paragraph declares its language and asks for automatic hyphens, which is why the words in these columns occasionally divide across lines. The result is a more even texture, with fewer rivers of white running down the column.

Columns introduce their own questions: where to break, whether a heading may sit at the foot of one, and how a quotation spanning both should interrupt the flow. Here the pull quote spans the full measure and the text resumes beneath it, balanced between the two columns.

Small text matters too. Notes, captions and legal lines are often set at seven or eight points, where hinting, stem darkening and subpixel positioning decide whether a reader squints. The footnotes below are set at seven and a half points in the same face as the body.[3]

Finally, notice the drop cap that opens this chapter. It is an ordinary letter, floated and enlarged, with the first line set in small capitals. A renderer need do nothing special for it, which is precisely the point: most typographic richness in a PDF is just careful placement of ordinary glyphs.

1 Word spacing in PDF is usually baked into glyph positions rather than expressed with the word-spacing operator, so readers rarely need to stretch anything themselves.

2 In a PDF a ligature is a single glyph; text extraction relies on the font's ToUnicode map to recover the letters it represents.

3 Try zooming to 400 percent here: well-rendered small text stays crisp and evenly weighted.

CHAPTER TWO

# OpenType *Features*

*Each row shows the same text with a feature off, then on. The glyphs differ; the underlying characters do not.*

| TAG | FEATURE | OFF | ON |
|---|---|---|---|
| liga | Standard ligatures | office fluffy fjord | office fluffy fjord |
| dlig | Discretionary ligatures | Asterisk & strict | Asterisk & strict |
| smcp | Small capitals | Printed in Boston | Printed in Boston |
| onum | Old-style figures | Since 1885: 3,467,920 | Since 1885: 3,467,920 |
| frac | Fractions | 1/2 cup, 3/4 inch, 7/8 | ½ cup, ¾ inch, ⅞ |
| swsh | Swash capitals | *The Quiet River* | *The Quiet River* |
| ss01 | Stylistic set 1 | a 1 4 6 9 G ? | a 1 4 6 9 G ? |
| kern | Kerning | AVATAR Type Wave | AVATAR Type Wave |
| tnum | Tabular figures | 1111 / 8888 | 1111 / 8888 |

## ONE VARIABLE FONT, NINE WEIGHTS

Ag 100 Ag 200 Ag 300 Ag 400 Ag 500 Ag 600 Ag 700 Ag 800 Ag 900

**System UI** · a single variable font file supplies every weight; the PDF embeds each instance as its own subset.

## SCRIPT FACES

*Handsome letters, joined in flight*

*Quietly Ornamental — 1234567890*

CHAPTER THREE

# Expressive *Type*

*Effects that become transparency groups, soft masks, clipping paths and transformed text objects once they reach PDF.*

Layered

**text-shadow** · two translucent offset copies composited under the glyphs

**Gradient-filled text** · an axial shading clipped to glyph outlines (text render mode 7)

Rotated

Skewed

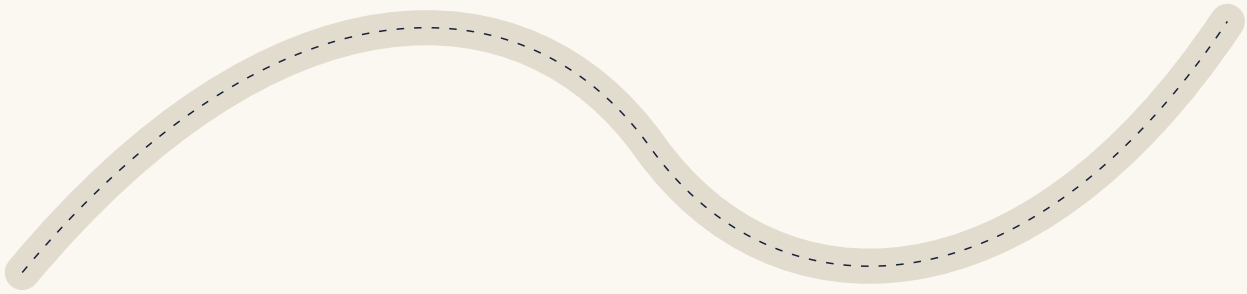
**transform** · skewX(−18°) on a filled box and rotate(−12°) on a label

Outline

TRACKED TO 600

**-webkit-text-stroke** · glyphs stroked, not filled (render mode 1)

## TEXT ON A CURVE



**SVG textPath** · each glyph is positioned and rotated individually along a cubic Bézier, filled with a gradient.

## HIERARCHY THROUGH CONTRAST

*Contrast is the* **engine** *of design.*

Pair a high-contrast Playfair italic with a heavy grotesque and the eye knows where to go. The PDF sees only two fonts and a handful of positioning operators; the hierarchy is entirely in the choices.

CHAPTER FOUR

# Scripts of the *World*

*A short greeting — roughly, "every letter deserves a good home" — in writing systems that need bidirectional layout, contextual shaping, reordering and vertical setting.*

ARABIC — RTL · JOINING
لكل حرف بيت
*Amiri*

HEBREW — RTL
לכל אות יש בית טוב
*Noto Sans Hebrew*

DEVANAGARI — CONJUNCTS
हर अक्षर का घर
*Noto Sans Devanagari*

THAI — NO SPACES
ทุกอักษรมีบ้าน
*Noto Sans Thai*

GREEK — POLYTONIC
Κάθε γράμμα ἔχει σπίτι
*EB Garamond*

CYRILLIC
Каждой букве — дом
*EB Garamond*

CHINESE — SIMPLIFIED
每个字都有家
*ZCOOL XiaoWei*

KOREAN — HANGUL
모든 글자에 집을
*Gowun Batang*

IPA — PHONETICS
/taɪp ɪz ˈkræft/
*International Phonetic Alphabet*

ARMENIAN
Ամեն տառին՝ տուն
*Noto Sans Armenian*

GEORGIAN
ყველა ასოს — სახლი
*Noto Sans Georgian*

TAMIL
எழுத்துக்கு வீடு
*Noto Sans Tamil*

JAPANESE
すべての文字に、
よい家を。
縦書きで読む。
*Shippori Mincho, writing-mode: vertical-rl*

## MIXED DIRECTION IN ONE LINE

The word كِتَاب means "book", and ספר does too — the bidi algorithm keeps each run in its own direction while the sentence stays left-to-right, even across numbers like ٣٤٥ and 345.

## COLOUR EMOJI

📄 🖋️ 🎨 📐 🔍 🌍 🧱 ✅ 🦀 📎

**Noto Color Emoji** (OFL) · colour bitmap glyphs that Chrome places into the PDF as images.

CHAPTER FIVE

# Mathematics

*Equations written in MathML Core and laid out by the browser: stretchy delimiters, radicals, large operators, scripts and matrices.*

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

**Quadratic formula** · fraction, radical, superscript

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

**Gaussian integral** · large operator with limits

$$e^{i\pi} + 1 = 0$$

**Euler's identity** · five constants, one line

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

**Basel problem** · summation with under- and over-scripts

$$\mathrm{R}\left(\theta\right) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

**Rotation matrix** · the transform behind every rotated page

$$B\left(t\right) = \sum_{i=0}^{3} \binom{3}{i} (1-t)^{3-i} t^i P_i$$

**Cubic Bézier** · the curve PDF paths are made of

## INLINE MATHEMATICS

A PDF page maps user space to device space with a matrix $\left[\,a\ b\ c\ d\ e\ f\,\right]$, so a point $\left(x, y\right)$ lands at $\left(ax + cy + e,\ bx + dy + f\right)$. With one point equal to $\frac{1}{72}$ inch, a Letter page measures $612 \times 792$ points, and a 300 dpi raster of it holds roughly $8.4 \times 10^{6}$ pixels.

## SYMBOLS AND ARROWS

$\forall\ \exists\ \varnothing\ \nabla\ \partial\ \in\ \notin\ \subset\ \subseteq\ \cup\ \cap\ \leq\ \geq\ \neq\ \approx\ \equiv\ \infty$

$\aleph\ \mathbb{R}\ \mathbb{C}\ \mathbb{Z}\ \mathbb{N}\ \rightarrow\ \Rightarrow\ \Leftrightarrow\ \mapsto\ \langle\ \rangle\ \lceil\ \rceil$

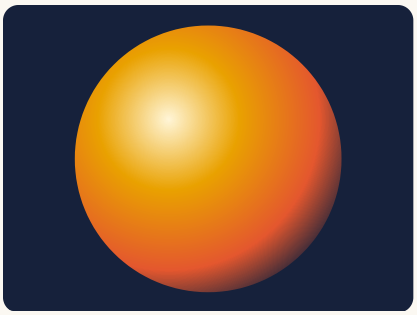
CHAPTER SIX

# Vector *Graphics*

*Nine small plates, each exercising one piece of the PDF imaging model.*



**Axial shading** · three-stop linear gradient



**Radial shading** · off-centre focal point



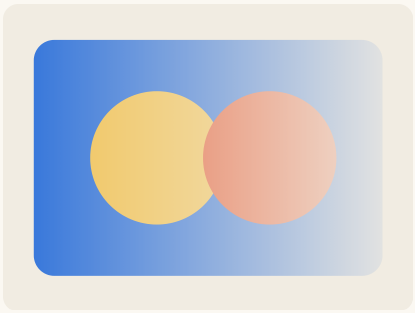
**Multiply** · subtractive overlaps, /BM /Multiply



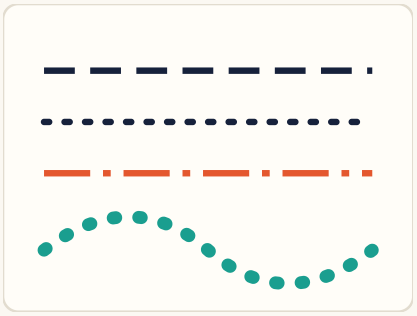
**Screen** · additive light, /BM /Screen



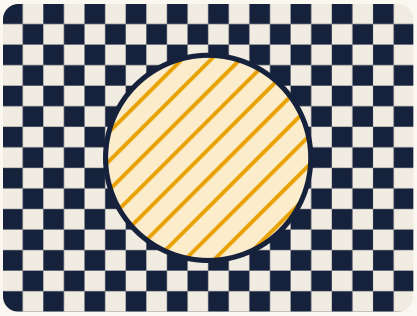

**Clipping path** · glyph outlines as a clip



**Soft mask** · luminosity gradient fades artwork



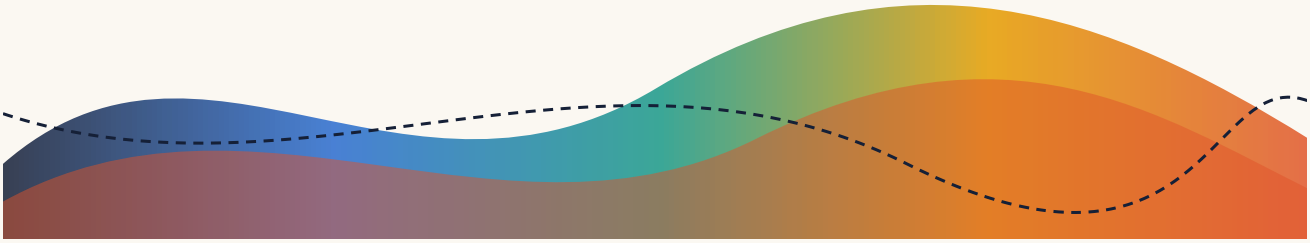
**Dash patterns** · caps, joins and phase



**Tiling patterns** · checks and rotated hatch



**Fill rules** · even-odd pentagram, 45% opacity



**Layered waves** · five-stop shading, 50% overlay in multiply mode, dashed stroke over the top.

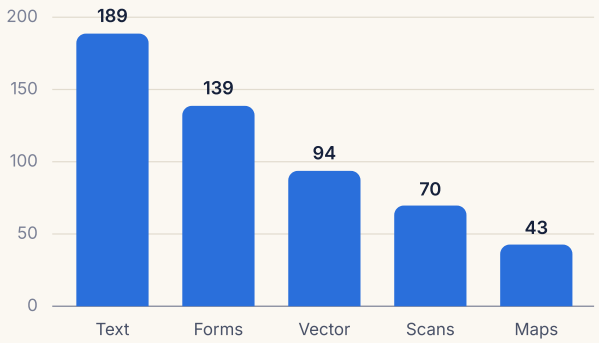
CHAPTER SEVEN

# Data & *Tables*

*Charts drawn as SVG and a striped table. All figures are illustrative sample data, not measurements.*

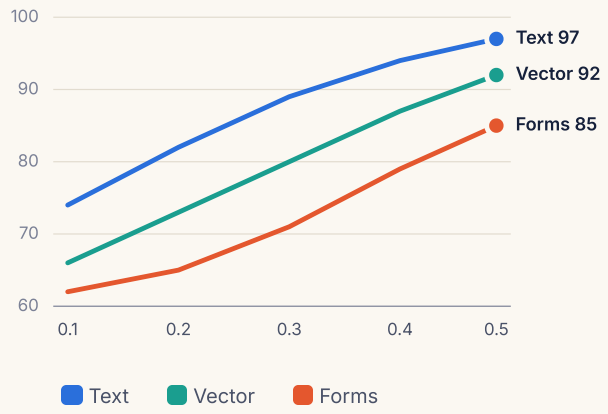
### Pages per second, by document type

Illustrative sample data · higher is better




### Renderer conformance over releases

Illustrative sample data · % of test pages matching reference




## WHERE EACH PDF FEATURE LIVES IN THIS FILE

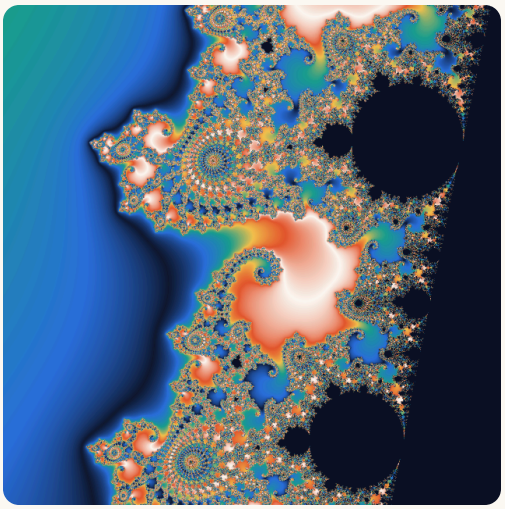
| FEATURE | PDF OBJECT | PRODUCED BY | PAGE | STATUS |
|---|---|---|---|---|
| Embedded font subsets | /FontFile2, /FontFile3 | Chrome | all | PRESENT |
| Tagged structure | /StructTreeRoot | Chrome | all | PRESENT |
| Axial & radial shadings | /ShadingType 2, 3 | Chrome | 6 | PRESENT |
| Blend modes | /ExtGState /BM | Chrome | 6 | PRESENT |
| Bookmarks | /Outlines | Chrome, xtask | — | PRESENT |
| Page labels | /PageLabels | xtask | — | PRESENT |
| Optional content | /OCProperties | xtask | 9, 10 | TWO LAYERS |
| Annotations | /Annots with /AP | xtask | 1, 9, 10 | PRESENT |
| Form fields | /AcroForm | xtask | 10 | PRESENT |
| File attachments | /EmbeddedFiles | xtask | — | PRESENT |
| Digital signature | /FT /Sig | xtask | 10 | UNSIGNED |

CHAPTER EIGHT

# Code & *Images*

*A monospaced listing with syntax colouring, and the raster that it describes.*

```rust
// Escape-time rendering of Seahorse Valley into an RGB buffer.
fn mandelbrot(width: u32, height: u32) -> Vec<u8> {
    let mut rgb = Vec::with_capacity((width * height * 3) as usize);
    for py in 0..height {
        for px in 0..width {
            let c = (CX + SPAN * (px as f64 / width as f64 - 0.5),
                     CY - SPAN * (py as f64 / height as f64 - 0.5));
            let (mut z, mut n) = ((0.0, 0.0), 0);
            while n < MAX_ITER && z.0 * z.0 + z.1 * z.1 <= 4.0 {
               z = (z.0 * z.0 - z.1 * z.1 + c.0, 2.0 * z.0 * z.1 + c.1);
               n += 1;
            }
            rgb.extend_from_slice(&palette(n)); // smooth colour ramp
        }
    }
    rgb
}
```



### A GENERATED RASTER

The image at left is computed by the xtask each time the showcase is built, encoded as PNG and handed to Chrome, which embeds it as an image XObject. Nothing is downloaded and no third-party artwork is involved, so the file stays reproducible and free to redistribute.

At 60 dpi the fractal edge looks smooth; zoom in and the individual pixels appear, a useful check that the viewer switches from filtered to nearest-neighbour sampling at the right moment.

### LINKS

- External: https://example.com/
- External: example.org/cleonpdf
- Internal: back to the contents or ahead to review & markup

CHAPTER NINE

# Review & *Markup*

*Every annotation on this page was added after printing, each with its own appearance stream, author, date and unique name.*

## TEXT MARKUP

Reviewers rarely rewrite a document; they highlight the phrase that matters, underline a claim that needs a source, and ~~strike out a redundant word~~ before moving on.

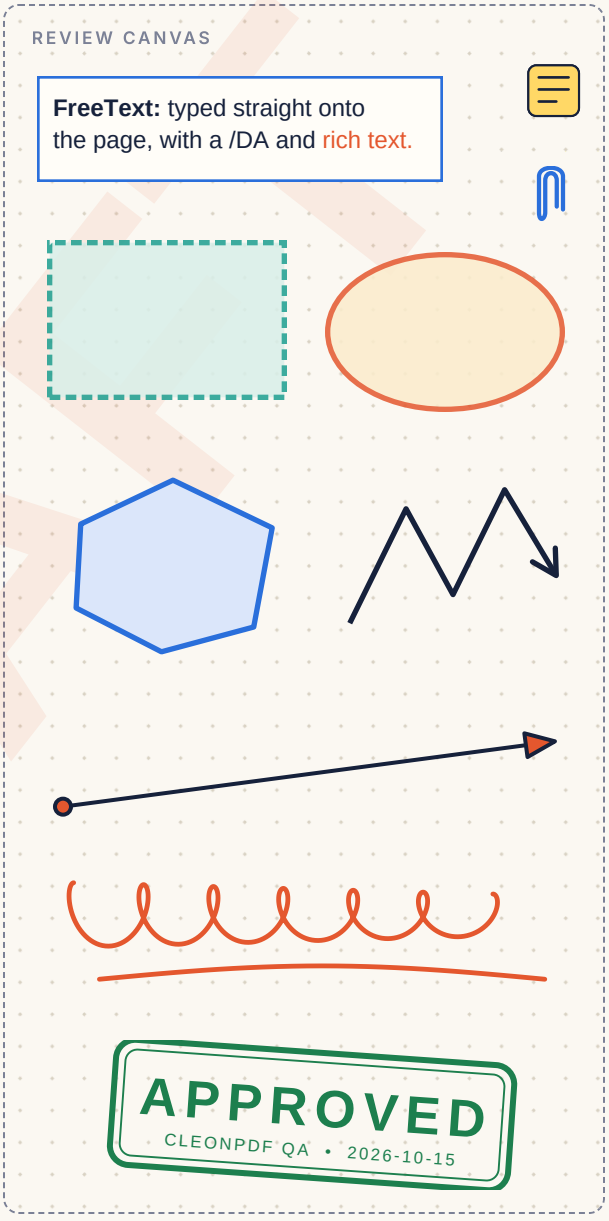
A wavy line says check the spelling here, while a caret marks where text should be inserted.

## COMMENTS

Sticky notes carry threaded replies, and a free-text box writes directly on the page. The paperclip holds an attached file that can be saved from the comment itself.

## LEGEND

1. Highlight
2. Underline
3. StrikeOut
4. Squiggly
5. Caret
6. Text + Popup
7. FreeText
8. Line (arrow)
9. Square
10. Circle
11. Polygon
12. PolyLine
13. Ink
14. Stamp
15. FileAttachment
16. Link




DRAFT

CHAPTER TEN

# Interactive *Form*

*This page was not printed by Chrome: the xtask writes its content stream and an AcroForm directly. Every widget ships its own appearance streams.*

## CONTACT

FULL NAME
Ada Typesetter

EMAIL
ada@example.com

DATE (AFDATE_FORMATEX YYYY-MM-DD)
2026-09-30

AMOUNT (AFNUMBER_FORMAT, RIGHT-ALIGNED)
$1,234.50

COMMENTS (MULTILINE)
CleonPDF renders this field from its own appearance stream, so it looks the same in every viewer. Edit it to see your viewer regenerate the text.

## PREFERENCES

CHECKBOXES

- [x] Subscribe to release notes
- [x] I have read the licence
- [ ] Send me stickers

PREFERRED OUTPUT (RADIO GROUP)

- ( ) PDF/A - archival
- (•) PDF/UA - accessible
- ( ) PDF/X - print exchange

PAPER SIZE (COMBO BOX)
Letter

FEATURES OF INTEREST (MULTI-SELECT LIST)
Rendering
Forms
Annotations
Signatures

## ACTIONS & SIGNATURE

PUSH BUTTON (JAVASCRIPT ACTION)

**Run JavaScript: app.alert**

JavaScript: shows app.alert(...) in viewers that run document scripts.

SIGNATURE (UNSIGNED /SIG FIELD)

×

Sign here - unsigned signature field

Links

Back to the contents

https://example.com/